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EXP:7

SELECT with various clause – GROUP BY, HAVING, ORDER BY

GROUP BY - HAVING

1. Display the course description and number of students registered in each course

```
mysql> SELECT C.CourseDescription,
    ->        COUNT(R.RegNo) AS Number_of_Students
    -> FROM COURSE C
    -> JOIN REGISTER R
    -> ON C.CourseNo = R.CourseNo
    -> GROUP BY C.CourseDescription;
+-------------------+--------------------+
| CourseDescription | Number_of_Students |
+-------------------+--------------------+
| Statistics        |                  2 |
| Programming       |                  3 |
| Database          |                  1 |
+-------------------+--------------------+
3 rows in set (0.09 sec)
```

2. Display the number of courses registered by each student

```
mysql> SELECT S.RegNo,
    ->        S.Name,
    ->        COUNT(R.CourseNo) AS Number_of_Courses
    -> FROM STUDENT S
    -> JOIN REGISTER R
    -> ON S.RegNo = R.RegNo
    -> GROUP BY S.RegNo,S.Name;
+-------+-------+-------------------+
| RegNo | Name  | Number_of_Courses |
+-------+-------+-------------------+
|   101 | Ravi  |                 2 |
|   102 | Priya |                 2 |
|   103 | Arun  |                 2 |
+-------+-------+-------------------+
3 rows in set (0.01 sec)
```

HAVING Example

Display courses having more than one student.

```
mysql> SELECT CourseNo,
    ->        COUNT(RegNo) AS Total_Students
    -> FROM REGISTER
    -> GROUP BY CourseNo
    -> HAVING COUNT(RegNo) > 1;
+----------+----------------+
| CourseNo | Total_Students |
+----------+----------------+
| C001     |              2 |
| C002     |              3 |
+----------+----------------+
2 rows in set (0.06 sec)
```